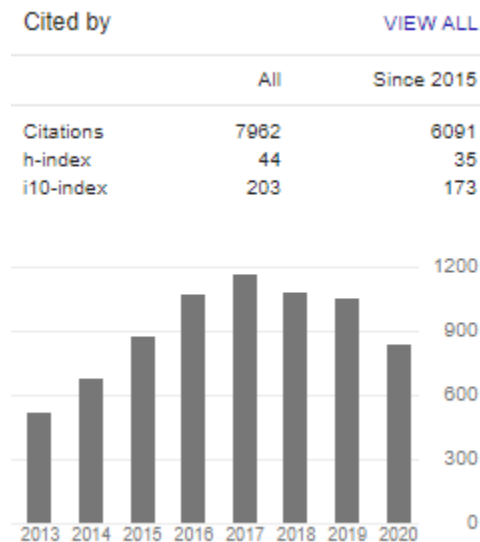


“Recent Trends In Cloud Computing Articles”

International Journal of Computer Science and Information Technology (IJCSIT)

Google Scholar Citation



ISSN: 0975-3826(online); 0975-4660 (Print)

<http://airccse.org/journal/ijcsit.html>

IOT SECURITY: PENETRATION TESTING OF WHITE-LABEL CLOUD-BASED IOT CAMERA COMPROMISING PERSONAL DATA PRIVACY

Marlon Intal Tayag, Francisco Napalit and Arcely Napalit, School of Computing Holy Angel University, Philippines

ABSTRACT

The Internet is driving force on how we communicate with one another, from posting messages and images to Facebook or “tweeting” your activities from your vacation. Today it is being used everywhere, now imagine a device that connects to the internet sends out data based on its sensors, this is the Internet-ofThings, a connection of objects with a plethora of sensors. Smart devices as they are commonly called, are invading our homes. With the proliferation of cheap Cloud-based IoT Camera use as a surveillance system to monitor our homes and loved ones right from the palm of our hand using our smartphones. These cameras are mostly white-label product, a process in which the product comes from a single manufacturer and bought by a different company where they are re-branded and sold with their own product name, a method commonly practice in the retail and manufacturing industry. Each Cloud-based IoT cameras sold are not properly tested for security. The problem arises when a hacker, hacks into the Cloud-based IoT Camera sees everything we do, without us knowing about it. Invading our personal digital privacy. This study focuses on the vulnerabilities found on White-label Cloud-based IoT Camera on the market specifically on a Chinese brand sold by Shenzhen Gwelltimes Technology. How this IoT device can be compromised and how to protect our selves from such cyber-attacks.

KEYWORDS

Network Protocols, Wireless Network, Mobile Network, Virus, Worms & Trojans, Internet of Things, Hacker, Smart Camera.

For More Details: <https://airconline.com/ijcsit/V12N5/12520ijcsit03.pdf>

Volume Link: http://airccse.org/journal/ijcsit2020_curr.html

REFERENCES

- [1] “PRIVACY | definition in the Cambridge English Dictionary.” [Online]. Available: <https://dictionary.cambridge.org/us/dictionary/english/privacy>. [Accessed: 23-Jun-2019].
- [2] “Summary: Philippines Data Privacy Act and implementing regulations.” [Online]. Available: <https://iapp.org/news/a/summary-philippines-data-protection-act-and-implementing-regulations/>. [Accessed: 23-Jun-2019].
- [3] D. G. Aneela, I. A. Anusha, K. Malavika, and R. Saripalle, “Research Trends of Network Security in IoT,” vol. 4863, no. September, pp. 6–10, 2017.
- [4] S. Ullah, L. Marcenaro, and B. Rinner, “Secure smart cameras by aggregate-signcryption with decryption fairness for multi-receiver IoT applications,” *Sensors (Switzerland)*, vol. 19, no. 2, 2019.
- [5] “The Ultimate Guide to White-Label Products & Solutions - Vendasta.” [Online]. Available: <https://www.vendasta.com/blog/the-ultimate-guide-to-white-label#how-white-label-works>. [Accessed: 09-Jul-2019].
- [6] “Why A White Label Solution Is Easier Than Building Your Own.” [Online]. Available: <https://www.forbes.com/sites/theyec/2014/06/03/why-a-white-label-solution-is-easier-than-building-your-own/#748a2186dd9e>. [Accessed: 09-Jul-2019].
- [7] “What is White Labeling? Pros and Cons of White Labeling Software | CallRail.” [Online]. Available: <https://www.callrail.com/blog/what-is-white-labeling/>. [Accessed: 24-Jun-2019].
- [8] K. Olha, “An investigation of lightweight cryptography and using the key derivation function for a hybrid scheme for security in IoT,” p. 42, 2017.
- [9] Y. Seralathan et al., “IoT security vulnerability: A case study of a Web camera,” *Int. Conf. Adv. Commun. Technol. ICACT*, vol. 2018-Febru, pp. 172–177, 2018.
- [10] J. Porras, J. Pänkäläinen, A. Knutas, and J. Khakurel, “Security In The Internet Of Things - A Systematic Mapping Study,” *Proc. 51st Hawaii Int. Conf. Syst. Sci.*, pp. 3750–3759, 2018.
- [11] J. Bugeja, D. Jönsson, and A. Jacobsson, “An Investigation of Vulnerabilities in Smart Connected Cameras,” 2018 IEEE Int. Conf. Pervasive Comput. Commun. Work. PerCom Work. 2018, pp. 537– 542, 2018. *International Journal of Computer Science & Information Technology (IJCSIT)* Vol 12, No 5, October 2020 40
- [12] R. Williams, E. McMahon, S. Samtani, M. Patton, and H. Chen, “Identifying vulnerabilities of consumer Internet of Things (IoT) devices: A scalable approach,” 2017 IEEE Int. Conf. Intell. Secur. Informatics Secur. Big Data, ISI 2017, pp. 179–181, 2017.
- [13] M. G. Samaila, M. Neto, D. A. B. Fernandes, M. M. Freire, and P. R. M. Inácio, “Security challenges of the Internet of Things,” *Internet of Things*, no. 9783319507569, pp. 53–82, 2017.
- [14] J. N. Goel and B. M. Mehtre, “Vulnerability Assessment & Penetration Testing as a Cyber Defence Technology,” *Procedia Comput. Sci.*, vol. 57, pp. 710–715, 2015.

- [15] “Angry IP Scanner - the original IP scanner for Windows, Mac and Linux.” [Online]. Available: <https://angryip.org/>. [Accessed: 10-Jul-2019].
- [16] “What is Nmap? Why you need this network mapper | Network World.” [Online]. Available: <https://www.networkworld.com/article/3296740/what-is-nmap-why-you-need-this-networkmapper.html>. [Accessed: 10-Jul-2019].
- [17] “What is Real Time Streaming Protocol (RTSP)? - Definition from Techopedia.” [Online]. Available: <https://www.techopedia.com/definition/4753/real-time-streaming-protocol-rtsp>. [Accessed: 10-Jul-2019].

AUTHORS

Dr. Marlon I. Tayag is a full-time Associate Professor at Holy Angel University and teaches Cyber Security subjects on Ethical Hacking and Forensic. He earned the degree of Doctor in Information Technology from St. Linus University in 2015 and is currently taking up Doctor of Philosophy in Computer Science at Technological Institute of the Philippines – Manila.



Dr. Tayag is Cisco Certified Network Associate, 210-250 CCNA Understanding Cisco Cybersecurity Fundamentals and Fluke CCTTA – Certified Cabling Test Technician Associate. Microsoft Certified Professional and Microsoft Certified Educator. Dr. Francisco D. Napalit, is a result driven IT professional who got a doctoral degree in Information Technology, with experience in the administration and support of information systems and network systems. Experienced in implementation, analysis, optimization, troubleshooting LAN/WAN network systems. Strong hands-on technical knowledge in CyberSec OPS, Cyber Crime Incident Response, MCP, CCNA, Fluke Networks certifications. Proven ability to lead and motivate project teams to ensure success. Track record for diagnosing complex problems and consistently delivering effective solutions. A solid 24 years work experience in different companies, institutions, organizations and currently the Dean of School of Computing at Holy Angel University. He is one of the founders and former vice president of Information Systems Security Association of the Philippines with direct experience in corporate and professional training, education and consulting in the field of I.T. and network systems. A subject matter expert (theoretical and practical), who got a hands-on experience in curriculum design and syllabus design in his varied work in different universities and colleges here and abroad. He is an individual who got strong business insight and passion for training and development, and with a good training and facilitation skills.



Prof. Arcely Perez-Napalit is a full-time faculty under the Computer Science Department of Holy Angel University. She's been teaching for almost two decades. One of her motto in teaching is to help students develop their logical and critical thinking and develop the character of a student as a whole. She also shared her passion for teaching overseas for six years. She is currently pursuing her postgraduate studies under the Ph.D. in Computer Science.



LIDAR POINT CLOUD CLASSIFICATION USING EXPECTATION MAXIMIZATION ALGORITHM

Nguyen Thi Huu Phuong

**Department of Software Engineering, Faculty of Information Technology, Hanoi
University of Mining and Geology, Hanoi, Vietnam**

ABSTRACT

EM algorithm is a common algorithm in data mining techniques. With the idea of using two iterations of E and M, the algorithm creates a model that can assign class labels to data points. In addition, EM not only optimizes the parameters of the model but also can predict device data during the iteration. Therefore, the paper focuses on researching and improving the EM algorithm to suit the LiDAR point cloud classification. Based on the idea of breaking point cloud and using the scheduling parameter for step E to help the algorithm converge faster with a shorter run time. The proposed algorithm is tested with measurement data set in Nghe An province, Vietnam for more than 92% accuracy and has faster runtime than the original EM algorithm.

KEYWORDS

LiDAR, EM algorithm, Scheduling parameter, LiDAR point elevation, GMM model

For More Details: <http://aircconline.com/ijcsit/V12N2/12220ijcsit01.pdf>

Volume Link: http://airccse.org/journal/ijcsit2020_curr.html

REFERENCES

- [1] Nallig Leal, Esmeide Leal, Sanchez Torres German [A linear programming approach for 3D point cloud simplification](#).
- [2] Zhenyang Hui, Dajung Li, Shuanggen Jin, Yao Yevenyo Ziggah, Leyang Wang (2019) Automatic DTM extraction from Airborne LiDAR based on expectation - maximization, Optics and Laser Technology, vol.112, pp. 43-55
- [3] Kun Zhang, Weihong Bi, Xiaoming Zhang, Xinghu Fu, Kunpeng Zhu, Li Zhu (2015) A new kmeans clustering algorithm for point cloud, International Journal of Hybrid Information Technology, vol. vol. 8, no. 9, pp. 157-170.
- [4] Keng FanLin, Chi Pei Wang, Pai Hui Su (2012) Object-based classification for LiDAR point cloud.
- [5] Chao Luo, Gunho Sohn (2013) [Line-based classification of terrestrial laser scanning data using conditional random field](#), International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol XL-7/W2, ISPRS2013.
- [6] Xiao Liu, Congyin Han, Tiande Guo (2018) A robust point sets matching method [Online]. Available: <https://arxiv.org/ftp/arxiv/papers/1411/1411.0791.pdf>. [Accessed 9 10 2019].
- [7] Z.Hui, P.Cheng, Y.Y. Ziggah, Y.Nie (2018) [A threshold-free filtering algorithm for airborne LiDAR point clouds based on Expectation Maximization](#), The International Archives of the Photogrammetry, RS and Spatial Information Sciences, vol. XLII-3.
- [8] Suresh Lodha, David P.Helmbold, Darren M.Fitzpatrick (2007) Aerial LiDAR data classification using expectation – maximization, Research Gate.
- [9] Yang HongLei, Peng JunHuan, Zhang DingXuan (2013) An Improved Em algorithm for remote sensing classification, Chinese Science Bullentin, vol. 58, no. 9, pp. 1060-1071.
- [10] Xiao Liu, Congying han, Tiande Guo (2014), "arXiv". [Online]. Available: <https://arxiv.org/ftp/arxiv/papers/1411/1411.0791.pdf>. [Accessed 7 10 2019].
- [11] Yu-chuan Chang, Ayman F.habib, Dong Cheon Lee, Jae Hong Yom (2008) Automatic classification of LiDAR data into Ground and non-Ground points, The International Archives of the Photogrammetry, RS and Spatial Information Sciences, vol. XXXVII, no. B4, pp.457-46.
- [12] Borja Rodriguez - Cuenca, Silverio Garcia Cortes, Celestino Ordonez, Maria C.Alonso (2015) [Automatic detection and classification of pole-like objects in urban point cloud data using an anomaly detection algorithm](#), Remote Sensing, vol. 7, pp.12680-12703.
- [13] Serez Kutluk, Koray Kayabol, Aydin Akan (2016) [Classification of Hyperspectral Images using Mixture of Probabilistics PCA models](#), European signal processing conference, pp. 1568-1572.
- [14] Iftekhar Naim, Daniel Gildea (2012) Convergence of EM algorithm for GMM with unbalanced Mixing coefficients, International Conference on Machine Learning.
- [15] Lawrence H. Cox, Marco Better (2009). Sampling from discrete distributions: Application to an editing problem, Research Gate.

- [16] Naonori Ueda, Ryohei Nakano (1998) Deterministic Annealing Variant of the EM algorithm, [Online]. Available: <https://papers.nips.cc/paper/941-deterministic-annealing-variant-of-the-em-algorithm.pdf>. [Accessed 6 3 2020].
- [17] S. Borman (2006) The expectation maximization algorithm a short tutorial.

AUTHOR

Born in 1985 in Ninh Binh province, Vietnam Country Graduated from the University of Mining and Geology University in 2008. Graduated Master of Science at the University of Natural Sciences - Vietnam National University, Hanoi in 2012. Author is currently a PhD student specialized in Information systems at the Institute of Information Technology, Vietnam Academy of Science and Technology. Currently working at: Faculty of Information Technology, University of Mining and Geology Research interests: Information System, Database, Data Mining, Geoinformatics



WEB SERVICES AS A SOLUTION FOR CLOUD ENTERPRISE RESOURCE PLANNING INTEROPERABILITY

Djamal Ziani and Nada Alfaadhel, King Saud University, Saudi Arabia

ABSTRACT

Recently, organizations have shown more interest in cloud computing because of the many advantages they provide (cost savings, storage capacity, scalability, and speed of loading). Enterprise resource planning (ERP) systems are one of the most important systems that have been upgraded to cloud computing. In this thesis, we focus on cloud ERP interoperability, which is an important challenge in cloud ERP. Interoperability is the ability of different components to work in independent clouds with no or minimum user effort. More than 20% of the risk rate of cloud adoption is caused by interoperability. Thus, we propose web services as a solution for cloud ERP interoperability. The proposed solution increases interoperability between different cloud service providers and between cloud ERP systems with other applications in a company.

KEYWORDS

Cloud computing, ERP, interoperability, web services

For More Details: <https://airconline.com/ijcsit/V12N1/12120ijcsit02.pdf>

Volume Link: http://airccse.org/journal/ijcsit2020_curr.html

REFERENCES

1. Purohit, G., M. Jaiswal, and S. Pandey, Challenges involved in implementation of ERP on demand solution: Cloud computing. *International Journal of Computer Science Issues (IJCSI)*, 2012. 9(4): p. 481.
2. Weng, F. and M.-C. Hung, Competition and challenge on adopting cloud ERP. *International Journal of Innovation, Management and Technology*, 2014. 5(4): p. 309.
3. Yang, N., D. Li, and Y. Tong, A cloud computing-based ERP system under the cloud manufacturing environment. *International Journal of Digital Content Technology and its Applications*, 2012. 6(23): p. 126.
4. Wang, S. and H. Wang, A Survey of Open Source Enterprise Resource Planning (ERP) Systems. *International Journal of Business and Information*, 2014. 9(1): p. 1.
5. AlDayel, A.I., M.S. Aldayel, and A.S. Al-Mudimigh, The critical success factors of ERP implementation in higher education in Saudi Arabia: a case study. *Journal of Information technology and economic development*, 2011. 2(2): p. 1.
6. Goyal, S., Public vs private vs hybrid vs community-cloud computing: A critical review. *International Journal of Computer Network and Information Security*, 2014. 6(3): p. 20.
7. Khanghahi, N., R. Nasiri, and M. Razavi, A New Approach toward Locating ERP Components on Cloud Computing Architecture. *International Journal of Advanced Research in Computer Science*, 2014. 5(1).
8. Xu, X., From cloud computing to cloud manufacturing. *Robotics and computer-integrated manufacturing*, 2012. 28(1): p. 75-86.
9. Sinjilawi, Y.K., M.Q. Al-Nabhan, and E.A. Abu-Shanab, Addressing security and privacy issues in cloud computing. *Journal of Emerging Technologies in Web Intelligence*, 2014. 6(2): p. 192-199.
10. Alali, F.A. and C.-L. Yeh, Cloud computing: Overview and risk analysis. *Journal of Information Systems*, 2012. 26(2): p. 13-33.
11. Fortinová, J., Risks of Cloud Computing. *Systémová Integrace*, 2013. 20(3).
12. Mather, T., S. Kumaraswamy, and S. Latif, Cloud security and privacy: an enterprise perspective on risks and compliance. 2009: " O'Reilly Media, Inc."
13. Mezgár, I. and U. Rauschecker, The challenge of networked enterprises for cloud computing interoperability. *Computers in Industry*, 2014. 65(4): p. 657-674. *International Journal of Computer Science & Information Technology (IJCSIT) Vol 12, No 1, February 2020* 40
14. Boillat, T. and C. Legner, From on-premise software to cloud services: the impact of cloud computing on enterprise software vendors' business models. *Journal of theoretical and applied electronic commerce research*, 2013. 8(3): p. 39-58.
15. Rimal, B.P., et al., Architectural requirements for cloud computing systems: an enterprise cloud approach. *Journal of Grid Computing*, 2011. 9(1): p. 3-26.

16. Suci, G., et al. ERP and e-business application deployment in open source distributed cloud systems. in The Eleventh International Conference on Informatics in Economy IE. 2012. Citeseer.
17. Mijac, M., R. Picek, and Z. Stapic. Cloud ERP system customization challenges. in Central European Conference on Information and Intelligent Systems. 2013. Faculty of Organization and Informatics Varazdin.
18. Gupta, V. and S.S. Bhatia, Developing Assurance Framework of Cloud Computing in the implementation of ERP: A Literature Survey.
19. Arunkumar, G. and N. Venkataraman, A novel approach to address interoperability concern in cloud computing. *Procedia Computer Science*, 2015. 50: p. 554-559.
20. Rezaei, R., T.K. Chiew, and S.P. Lee, A review on E-business Interoperability Frameworks. *Journal of Systems and Software*, 2014. 93: p. 199-216.
21. Yu, Q., L. Chen, and B. Li, Ant colony optimization applied to web service compositions in cloud computing. *Computers & Electrical Engineering*, 2015. 41: p. 18-27.
22. Nacer, H. and D. Aissani, Semantic web services: Standards, applications, challenges and solutions. *Journal of Network and Computer Applications*, 2014. 44: p. 134-151.
23. Cabral, L., et al., Approaches to semantic web services: an overview and comparisons, in *The semantic web: Research and applications*. 2004, Springer. p. 225-239.
24. Sheng, Q.Z., et al., Web services composition: A decade's overview. *Information Sciences*, 2014. 280: p. 218-238.
25. Bertolino, A. and A. Polini. The audition framework for testing web services interoperability. in *Software Engineering and Advanced Applications*, 2005. 31st EUROMICRO Conference on. 2005. IEEE.
26. Curbera, F., W. Nagy, and S. Weerawarana. Web services: Why and how. in *Workshop on ObjectOriented Web Services-OOPSLA*. 2001.
27. Avram, M.-G., Advantages and challenges of adopting cloud computing from an enterprise perspective. *Procedia Technology*, 2014. 12: p. 529-534.
28. Panetto, H., et al., New perspectives for the future interoperable enterprise systems. *Computers in Industry*, 2015.
29. Romero, D. and F. Vernadat, Enterprise information systems state of the art: Past, present and future trends. *Computers in Industry*, 2016.
30. Hanna, S., An Approach to Modeling Web Services Datatype Descriptions. *Journal of Theoretical and Applied Electronic Commerce Research*, 2016. 11(2): p. 64. *International Journal of Computer Science & Information Technology (IJCSIT)* Vol 12, No 1, February 2020 41
31. Narock, T., V. Yoon, and S. March, A provenance-based approach to semantic web service description and discovery. *Decision Support Systems*, 2014. 64: p. 90-99. 32. Tsalgatidou, A. and T. Pilioura, An overview of standards and related technology in web services. *Distributed and Parallel Databases*, 2002. 12(2-3): p. 135-162.

33. Metin, S. Using Web Services for WebRTC signaling interoperability. in Network Operations and Management Symposium (NOMS), 2016 IEEE/IFIP. 2016. IEEE.
34. Danila, C., et al., Web-service based architecture to support SCM context-awareness and interoperability. Journal of Intelligent Manufacturing, 2016. 27(1): p. 73-82. 35. Bhukya, D.P., R.A. Sony, and G. Muduganti, On web services based cloud interoperability. International Journal of Computer Science Issues (IJCSI), 2012. 9(5): p. 232

BIG DATA IN CLOUD COMPUTING REVIEW AND OPPORTUNITIES

Manoj Muniswamaiah, Tilak Agerwala and Charles Tappert
Seidenberg School of CSIS, Pace University, White Plains, New York

ABSTRACT

Big Data is used in decision making process to gain useful insights hidden in the data for business and engineering. At the same time it presents challenges in processing, cloud computing has helped in advancement of big data by providing computational, networking and storage capacity. This paper presents the review, opportunities and challenges of transforming big data using cloud computing resources.

KEYWORDS

Big data; cloud computing; analytics; database; data warehouse

For More Details: <http://aircconline.com/ijcsit/V11N4/11419ijcsit04.pdf>

Volume Link: http://airccse.org/journal/ijcsit2019_curr.html

REFERENCES

- [1] Konstantinou, I., Angelou, E., Boumpouka, C., Tsoumakos, D., & Koziris, N. (2011, October). [On the elasticity of nosql databases over cloud management platforms](#). In Proceedings of the 20th ACM international conference on Information and knowledge management (pp. 2385-2388). ACM.
- [2] Labrinidis, Alexandros, and Hosagrahar V. Jagadish. "[Challenges and opportunities with big data](#)." Proceedings of the VLDB Endowment 5.12 (2012): 2032-2033.
- [3] Abadi, D. J. (2009). Data management in the cloud: Limitations and opportunities. IEEE Data Eng. Bull, 32(1), 3-12.
- [4] Luhn, H. P. (1958). A business intelligence system. IBM Journal of Research and Development, 2(4), 314-319 International Journal of Computer Science & Information Technology (IJCSIT) Vol 11, No 4, August 2019 57
- [5] Sivarajah, Uthayasankar, et al. "[Critical analysis of Big Data challenges and analytical methods](#)." Journal of Business Research 70 (2017): 263-286.
- [6] <https://www.bmc.com/blogs/saas-vs-paas-vs-iaas-whats-the-difference-and-how-to-choose/>
- [7] Kavis, Michael J. Architecting the cloud: design decisions for cloud computing service models (SaaS, PaaS, and IaaS). John Wiley & Sons, 2014.
- [8] https://www.ripublication.com/ijaer17/ijaerv12n17_89.pdf
- [9] Sakr, S. & Gaber, M.M., 2014. Large Scale and big data: Processing and Management Auerbach, ed.
- [10] Ji, Changqing, et al. "Big data processing in cloud computing environments." 2012 12th international symposium on pervasive systems, algorithms and networks. IEEE, 2012.
- [11] Han, J., Haihong, E., Le, G., & Du, J. (2011, October). Survey on nosql database. In Pervasive Computing and Applications (ICPCA), 2011 6th International Conference on (pp. 363-366). IEEE.
- [12] Zhang, L. et al., 2013. Moving big data to the cloud. INFOCOM, 2013 Proceedings IEEE, pp.405-409
- [13] Fernández, Alberto, et al. "[Big Data with Cloud Computing: an insight on the computing environment, MapReduce, and programming frameworks](#)." Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery 4.5 (2014): 380-409.
- [14] http://acme.able.cs.cmu.edu/pubs/uploads/pdf/IoTBD_2016_10.pdf
- [15] Xiaofeng, Meng, and Chi Xiang. "[Big data management: concepts, techniques and challenges \[J\]](#)." Journal of computer research and development 1.98 (2013): 146-169.
- [16] Muniswamaiah, Manoj & Agerwala, Tilak & Tappert, Charles. (2019). [Challenges of Big Data Applications in Cloud Computing](#). 221-232. 10.5121/csit.2019.90918.

SMART MOTORCYCLE HELMET: REAL-TIME CRASH DETECTION WITH EMERGENCY NOTIFICATION, TRACKER AND ANTI-THEFT SYSTEM USING INTERNET-OF-THINGS CLOUD BASED TECHNOLOGY

Marlon Intal Tayag¹ and Maria Emmalyn Asuncion De Vigal Capuno²

¹College of Information and Communications Technology Holy Angel University, Angeles, Philippines

²Faculty of Information Technology Future University, Khartoum, Sudan

ABSTRACT

Buying a car entails a cost, not counting the day to day high price tag of gasoline. People are looking for viable means of transportation that is cost-effective and can move its way through traffic faster. In the Philippines, motorcycle was the answer to most people transportation needs. With the increasing number of a motorcycle rider in the Philippines safety is the utmost concern. Today technology plays a huge role on how this safety can be assured. We now see advances in connected devices. Devices can sense its surrounding through sensor attach to it. With this in mind, this study focuses on the development of a wearable device named Smart Motorcycle Helmet or simply Smart Helmet, whose main objective is to help motorcycle rider in times of emergency. Utilizing sensors such as alcohol level detector, crash/impact sensor, Internet connection thru 3G, accelerometer, Short Message Service (SMS) and cloud computing infrastructure connected to a Raspberry Pi Zero-W and integrating a separate Arduino board for the anti-theft tracking module is used to develop the propose Internet-of Things (IoT) device.

Using quantitative method and descriptive type research, the researchers validated the results from the inputs of the participant who tested the smart helmet during the alpha and beta testing process. Taking into account the ethical consideration of the volunteers, who will test the Smart Helmet. To ensure the reliability of the beta and alpha testing, ISO 25010 quality model was used for the assessment focusing on the device accuracy, efficiency and functionality. Based on the inputs and results gathered, the proposed Smart Helmet IoT device can be used as a tool in helping a motorcycle rider when an accident happens to inform the first-responder of the accident location and informing the family of the motorcycle rider.

KEYWORDS

Smart Helmet, Internet of Things, Sensors, Real-Time Crash Detection, Emergency Notification, Tracker, Anti-Theft System Cloud Based Technology

For More Details: <http://aircconline.com/ijcsit/V11N3/11319ijcsit07.pdf>

Volume Link: http://airccse.org/journal/ijcsit2019_curr.html

REFERENCES

- [1] Mascarinas, E. M. (2016). Study in better safety measures for motorcycles urged - SUNSTAR. Retrieved December 11, 2018, from <https://www.sunstar.com.ph/article/111646>
- [2] L. Ramos. (2018). Road Accidents In The Philippines: Key Figures - eCompareMo - eCompareMo. Retrieved December 11, 2018, from <https://www.ecomparemo.com/info/road-accidents-in-the-philippines-key-figures/>
- [3] Nandu, R., & Singh, K. (2014). Smart Helmet for Two-Wheelers. *Advances in Automobile Engineering*, 03(01), 1–2. <https://doi.org/10.4172/2167-7670.1000110>
- [4] Sung, K.-M., Noble, J., Kim, S.-C., Jeon, H.-J., Kim, J.-Y., Do, H.-H., ... Baek, K.-J. (2016). The Preventive Effect of Head Injury by Helmet Type in Motorcycle Crashes: A Rural Korean Single-Center Observational Study. *BioMed Research International*, 2016, 1–7. <https://doi.org/10.1155/2016/1849134>
- [5] J. Dodson. (n.d.). Motorcycle Crashes and Brain Injuries | Jim Dodson Law. Retrieved December 11, 2018, from <https://www.jimdodsonlaw.com/library/motorcycle-crashes-and-brain-injuries.cfm>
- [6] W. Tan. (2018). WHO PH: Over 90% of Motorcycle Deaths Didn't Wear Helmets - Carmudi Philippines. Retrieved December 18, 2018, from <https://www.carmudi.com.ph/journal/philippines-90-motorcycle-deaths-didnt-wear-helmets/>
- [7] Lahaussé, J. A., Fildes, B. N., Page, Y., & Fitzharris, M. P. (2008). The potential for automatic crash notification systems to reduce road fatalities. *Annals of Advances in Automotive Medicine. Association for the Advancement of Automotive Medicine. Annual Scientific Conference*, 52, 85–92. Retrieved from <http://www.ncbi.nlm.nih.gov/pubmed/19026225>
- [8] M. Glasscoe. (n.d.). What is GPS? Retrieved December 11, 2018, from <http://sceinfo.usc.edu/education/k12/learn/gps1.htm>
- [9] Brodsky, H. (n.d.). *Emergency Medical Service Rescue Time in Fatal Road Accidents*. Retrieved from <http://onlinepubs.trb.org/Onlinepubs/trr/1990/1270/1270-011.pdf>
- [10] J. Porter. (2018). The History of IoT (Internet of Things) and How It's Changed Today. Retrieved December 18, 2018, from <https://www.techprevue.com/history-iot-changed-today/>
- [11] Vishal, D., Afaque, H. S., Bhardawaj, H., & Ramesh, T. K. (2018). IoT-driven road safety system. *International Conference on Electrical, Electronics, Communication Computer Technologies and Optimization Techniques, ICEECOT 2017, 2018-Janua*, 862–866. <https://doi.org/10.1109/ICEECOT.2017.8284624>
- [12] Dickenson. (2016). How IoT and machine learning can make our roads safer | TechCrunch. Retrieved December 18, 2018, from <https://techcrunch.com/2016/07/13/how-iot-and-machine-learning-can-make-our-roads-safer/>

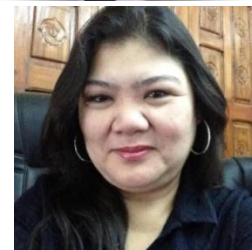
- [13] Faizan Manzoor, S. A. B. (2017). Faaz smart helmet, 6(6), 332–335.
- [14] Hobby, K. C., Gowing, B., & Matt, D. P. (2016). Smart helmet, 5(3), 660–663.
- [15] Khaja, M., Aatif, A., & Manoj, A. (2017). [Smart Helmet Based On IoT Technology](#), 5(Vii), 409–413.
- [16] Motorcyclist age group and gender data - TAC - Transport Accident Commission. (n.d.). Retrieved April 20, 2019, from <http://www.tac.vic.gov.au/road-safety/statistics/summaries/motorcycle-crash-data/motorcyclist-age-group-and-gender-data>
- [17] Amir, G. (n.d.). Prototyping Model in Software Development and Testing. Retrieved January 22, 2019, from <https://www.testingexcellence.com/prototyping-model-software-development/>
- [18] The Importance of Alpha & Beta Testing Services | Software Testing Tips and Best Practices. (n.d.). Retrieved April 21, 2019, from <https://blog.testmatick.com/2016/04/19/the-importance-of-alpha-beta-testing-services/>

AUTHORS

Dr. Marlon I. Tayag is a full-time Associate Professor at Holy Angel University and teaches Cyber Security subjects on Ethical Hacking and Forensic. He earned the degree of Doctor in Information Technology from St. Linus University in 2015 and is currently taking up Doctor of Philosophy in Computer Science at Technological Institute of the Philippines – Manila. Dr. Tayag is Cisco Certified Network Associate, 210-250 CCNA Understanding Cisco Cybersecurity Fundamentals and Fluke CCTTA – Certified Cabling Test Technician Associate. Microsoft Certified Professional and Microsoft Certified Educator.



Dr. Ma. Emmalyn A. V. Capuno is a currently the Dean of the Faculty of Information Technology of Future University Sudan with the academic rank of Associate Professor; a position she has been holding since 2009. She earned the degree of Doctor of Philosophy in Information Technology Management from Colegio de San Juan Letran – Calamba, Philippines in 2005. Her teaching and research expertise includes Operating Systems, Knowledge Management, Business Intelligence and many more.



A SURVEY ON SECURITY CHALLENGES OF VIRTUALIZATION TECHNOLOGY IN CLOUD COMPUTING

Nadiah M. Almutairy¹ and Khalil H. A. Al-Shqeerat²

¹Computer Science Department, College of Sciences and Arts in Rass, Saudi Arabia

²Computer Science Department, Qassim University, Saudi Arabia

ABSTRACT

Virtualization has become a widely and attractive employed technology in cloud computing environments. Sharing of a single physical machine between multiple isolated virtual machines leading to a more optimized hardware usage, as well as make the migration and management of a virtual system more efficiently than its physical counterpart. Virtualization is a fundamental technology in a cloud environment. However, the presence of an additional abstraction layer among software and hardware causes new security issues. Security issues related to virtualization technology have become a significant concern for organizations due to arising some new security challenges.

This paper aims to identify the main challenges and risks of virtualization in cloud computing environments. Furthermore, it focuses on some common virtual-related threats and attacks affect the security of cloud computing.

The survey was conducted to obtain the views of the cloud stakeholders on virtualization vulnerabilities, threats, and approaches that can be used to overcome them.

Finally, we propose recommendations for improving security, and mitigating risks encounter virtualization that necessary to adopt secure cloud computing.

KEYWORDS

Cloud Computing, Virtualization, Security, Challenge, Risk

For More Details: <http://airconline.com/ijcsit/V11N3/11319ijcsit08.pdf>

Volume Link: http://airccse.org/journal/ijcsit2019_curr.html

REFERENCES

- [1] B. Loganayagi and S. Sujatha, "Creating virtual platform for cloud computing," in Proc. 2010 IEEE International Conference on Computational Intelligence and Computing Research (ICCIC), 2010, pp. 1-4.
- [2] L. Garber, "[The Challenges of Securing the Virtualized Environment](#)," Computer, vol. 45, no. 1, pp. 17-20, 2012.
- [3] Cloud Security Alliance, "Top threats to cloud computing V1.0," CSA, 2010. [Online]. Available: <https://cloudsecurityalliance.org/topthreats/csathreats.v1.0.pdf>. [Accessed: Nov.-2017].
- [4] Cloud Security Alliance, "The Notorious Nine. Cloud Computing Top Threats in 2013," CSA, 2013. [Online]. Available: <http://www.cloudsecurityalliance.org/topthreats.%5Cnhttp://www.cloudsecurityalliance.org>. [Accessed: Oct.-2017].
- [5] G.Xiaopeng, W.Sumei, and C.Xianqin, "VNSS: A network security sandbox for virtual computing environment," In Proc. 2010 IEEE Youth Conference on Information, Computing and Telecommunications, 2010, pp. 395–398.
- [6] N. Afshan, "[Analysis and Assessment of the Vulnerabilities in Cloud Computing](#)," Int. J. Adv. Res. Comput. Sci., vol. 8, no. 2, 2017, pp. 2015–2018.
- [7] S. Bulusu and K. Sudia, "A Study on Cloud Computing Security Challenges," Master thesis, School of Computing at Blekinge Institute of Technology, 2012.
- [8] H. Wu, Y. Ding, C. Winer, and L. Yao, "[Network Security for Virtual Machine in Cloud Computing](#)," in Proc. 5th International Conference on Computer Sciences and Convergence Information Technology, 2009, pp. 18–21.
- [9] M. R. Anala, J. Shetty, and G. Shobha, "A framework for secure live migration of virtual machines," in Proc. Int. Conf. Adv. Comput. Commun. Informatics, ICACCI 2013, 2013, pp. 243–248.
- [10] A. Parashar and A. Borde, "[Cloud Computing: Security Issues and its Detection Methods](#)," Int. J. of Engg. Sci. & Mgmt., vol. 5, no. 2, 2015, pp. 136–140.
- [11] J. Wei, X. Zhang, G. Ammons, V. Bala, and P. Ning, "[Managing security of virtual machine images in a cloud environment](#)," in Proc. ACM workshop on Cloud computing security - CCSW '09, 2009, p. 91.

- [12] Cloud Security Alliance, “Best Practices for Mitigating Risks in Virtualized Environments,” Downloads.cloudsecurityalliance.org, 2015. [Online]. Available: [https://downloads.cloudsecurityalliance.org/whitepapers/Best Practices for%20 Mitigating Risks Virtual Environments_April2015_4-1-15_GLM5.pdf](https://downloads.cloudsecurityalliance.org/whitepapers/Best_Practices_for%20Mitigating_Risks_Virtual_Environments_April2015_4-1-15_GLM5.pdf). [Accessed: 11- Jan- 2017].
- [13] T. Garfinkel and M. Rosenblum, “[When Virtual is Harder Than Real: Security Challenges in Virtual Machine Based Computing Environments](#),” in Proc. 10th Conf. Hot Top. Oper. Syst., 2005, pp. 20–25.
- [14] I. Studnia, E. Alata, Y. Deswarte, M. Kaâniche, and V. Nicomette, “[Survey of Security Problems in Cloud Computing Virtual Machines](#),” in Proc. Computer and Electronics Security Applications Rendez-vous (C&ESAR), 2012, pp. 61–74.
- [15] Y. Azar, S. Kamara, I. Menache, M. Raykova, and B. Shepherd, “[Co-Location-Resistant Clouds](#),” in Proc. 6th edition of the ACM Workshop on Cloud Computing Security, 2014, pp. 9-20.
- [16] V. Varadarajan, “[Isolation in Public Clouds: Threats, Challenges and Defenses](#)”, PhD thesis, University of Wisconsin–Madison, 2015.
- [17] I. Khalil, A. Khreishah, and M. Azeem, “[Cloud Computing Security: A Survey](#),” Computers, vol. 3, no. 1, 2014, pp. 1–35.
- [18] Y. Xia, Y. Liu, H. Chen, and B. Zang, “[Defending against VM rollback attack](#),” in Proc. IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN 2012), 2012, pp. 1-5.
- [19] R.D. Londhe and S.S. Sherekar, “[Imperial Analysis of Threats and Vulnerabilities in Cloud Computing](#),” International Journal of Advanced Research in Computer Science, vol. 5, no. 4 2014, pp. 12-17.
- [20] T. Brooks, C. Caicedo, and J. Park, “[Security challenges and countermeasures for trusted virtualized computing environments](#),” In Proc. World Congress on Internet Security (WorldCIS-2012), 2012, pp. 117 – 122.
- [21] S. Luo, Z. Lin, X. Chen, Z. Yang, and J. Chen, “[Virtualization security for cloud computing service](#),” in Proc. Int. Conf. Cloud Serv. Comput., 2011, pp. 174–179.
- [22] R. Schwarzkopf, (2015) “Virtual Machine Lifecycle Management in Grid and Cloud computing,” University of Marburg. [Online]. Available: <http://archiv.ub.uni-marburg.de/diss/z2015/0407/pdf/drs.pdf>. [Accessed: Aug.-2017].

- [23] V.K. Veeramachaneni, "Security Issues and Countermeasures in Cloud Computing Environment," Int. J. Eng. Sci. Innov. Technol., vol. 4, no. 5, 2015, pp. 82–93.
- [24] K. Owens, "Securing Virtual Compute Infrastructure in the Cloud," SAVVIS, 2009.[Online]. Available:
http://viewer.media.bitpipe.com/1018468865_999/1296679360_880/Securing-Virtual-Compute-Infrastructure-in-the-Cloud.pdf. [Accessed: Jun-2017].
- [25] S. Z. I. Tariqul and D. Manivannan, "[A Classification and Characterization of Security Threats in Cloud Computing](#)," Int. J. Next-Generation Comput., vol. 7, no. 1, pp. 1–17, 2016.
- [26] C. N. Modi and K. Acha, "[Virtualization layer security challenges and intrusion detection/prevention systems in cloud computing: a comprehensive review](#)," J. Supercomput., vol. 73, no. 3, 2017, pp. 1192-1234.

GROUP BASED RESOURCE MANAGEMENT AND PRICING MODEL IN CLOUD COMPUTING

Shelia Rahman, Afroza Sultana , Afsana Islam, and Md Whaiduzzaman

Institute Of Information Technology,JahangirnagarUniversity,Dhaka,Bangladesh.

ABSTRACT

Cloud computing utilizes large scale computing infrastructure that has been radically changing the IT landscape enabling remote access to computing resources with low service cost, high scalability , availability and accessibility. Serving tasks from multiple users where the tasks are of different characteristics with variation in the requirement of computing power may cause under or over utilization of resources. Therefore maintaining such mega-scale datacenter requires efficient resource management procedure to increase resource utilization. However, while maintaining efficiency in service provisioning it is necessary to ensure the maximization of profit for the cloud providers. Most of the current research works aims at how providers can offer efficient service provisioning to the user and improving system performance. There are comparatively fewer specific works regarding resource management which also deals with the economic section that considers profit maximization for the provider. In this paper we represent a model that deals with both efficient resource utilization and pricing of the resources. The joint resource management model combines the work of user assignment, task scheduling and load balancing on the fact of CPU power endorsement. We propose four algorithms respectively for user assignment, task scheduling, load balancing and pricing that works on group based resources offering reduction in task execution time(56.3%),activated physical machines(41.44%),provisioning cost(23%) . The cost is calculated over a time interval involving the number of served customer at this time and the amount of resources used within this time.

KEYWORDS

Resource Management, Resource Pricing, Task Execution, Load Balancing, Task Scheduling.

For More Details: <http://aircconline.com/ijcsit/V10N4/10418ijcsit03.pdf>

Volume Link: http://airccse.org/journal/ijcsit2018_curr.html

REFERENCES

- [1] R. Buyya, C. S. Yeo, S. Venugopal, J. Broberg, I. Brandic, Cloud computing and emerging platforms: Vision, hype, and reality for delivering computing as the 5th utility, *Future Generation computer systems* 25 (2009) 599–616..
- [1] M. Whaiduzzaman, M. N. Haque, M. RejaulKarimChowdhury, A. Gani, A study on strategic provisioning of cloud computing services, *The Scientific World Journal* 2014 (2014)
- [2] M. Alba, IoT Devices to Outnumber Humans in 2017, <https://www.engineering.com/IOT/ArticleID/15594/IoT-Devices-to-Outnumber-Humans-in2017.aspx>, 2017. [Online; accessed 8-May-2018].
- [3] S. Akter, M. Whaiduzzaman, Dynamic service level agreement verification in cloud computing, *IJCSIS* (2017).
- [4] N. C. Luong, P. Wang, D. Niyato, Y. Wen, Z. Han, Resource management in cloud networking using economic analysis and pricing models: A survey, *IEEE Communications Surveys & Tutorials* 19 (2017) 954–1001.
- [5] R. weber, Cost Based Pricing, <https://onlinelibrary.wiley.com/doi/abs/10.1002/0470867175.ch7>, 2003. [Online; accessed 9-May-2018]. [6] K. H. Prasad, T. A. Faruque, L. V. Subramaniam, M. Mohania, G. Venkatachaliah, Resource allocation and sla determination for large data processing services over cloud, in: *Services Computing (SCC), 2010 IEEE International Conference on*, IEEE, pp. 522–529.
- [7] D. Di Spaltro, A. Polvi, L. Welliver, Methods and systems for cloud computing management, 2016. US Patent 9,501,329.
- [8] M. Shojafar, N. Cordeschi, E. Baccarelli, Energy-efficient adaptive resource management for realtime vehicular cloud services, *IEEE Transactions on Cloud computing* (2016)
- [9] E. Oppong, S. Khaddaj, H. E. Elastriss, Cloud computing: resource management and service allocation, in: *Distributed Computing and Applications to Business, Engineering & Science (DCABES), 2013 12th International Symposium on*, IEEE, pp. 142–145.
- [10] D. Ajmire, M. Atique, Grouping based load balancing in cloud computing, *International Journal of Innovative Research and Development* 5 (2016). [11] S. Abrishami, M. Naghibzadeh, D. H. Epema, Deadline-constrained workflow scheduling algorithms for infrastructure as a service clouds, *Future Generation Computer Systems* 29 (2013) 158–169.
- [12] Z. Tang, L. Qi, Z. Cheng, K. Li, S. U. Khan, K. Li, An energy-efficient task scheduling algorithm in dvfs-enabled cloud environment, *Journal of Grid Computing* 14 (2016) 55–74.
- [13] J. M. Galloway, K. L. Smith, S. S. Vrbsky, Power aware load balancing for cloud computing, in: *Proceedings of the World Congress on Engineering and Computer Science*, volume 1, pp. 19–21.
- [14] E. Ibrahim, N. A. El-Bahnasawy, F. A. Omara, Task scheduling algorithm in cloud computing environment based on cloud pricing models, in: *Computer Applications & Research (WSCAR), 2016 World Symposium on*, IEEE, pp. 65–71.
- [15] H. K. Ala'a Al-Shaikh, A. Sharieh, A. Sleit, Resource utilization in cloud computing as an optimization problem, *Resource* 7 (2016).

- [16] M. Whaiduzzaman, A. Naveed, A. Gani, Mobicore: Mobile device based cloudlet resource enhancement for optimal task response, *IEEE Transactions on Services Computing* (2016).
- [17] Whaiduzzaman, Md. "Performance enhancement framework for cloudlet in mobile cloud computing." PhD diss., Fakulti Sains Komputer dan Teknologi Maklumat, Universiti Malaya, 2016.
- [18] Whaiduzzaman, Md, Abdullah Gani, and Anjum Naveed. "Pefc: performance enhancement framework for cloudlet in mobile cloud computing." In *Robotics and Manufacturing Automation (ROMA)*, 2014 IEEE International Symposium on, pp. 224-229. IEEE, 2014.
- [19] N. Zhang, H. Hämmäinen, Cost efficiency of sdn in lte-based mobile networks: Case finland, in: *Networked Systems (NetSys)*, 2015 International Conference and Workshops on, IEEE, pp. 1–5
- [20] Z. Cao, J. Lin, C. Wan, Y. Song, Y. Zhang, X. Wang, Optimal cloud computing resource allocation for demand side management in smart grid, *IEEE Transactions on Smart Grid* 8 (2017) 1943–1955.
- [21] K. Tsakalozos, H. Kllapi, E. Sitaridi, M. Roussopoulos, D. Paparas, A. Delis, Flexible use of cloud resources through profit maximization and price discrimination, in: *Data Engineering (ICDE)*, 2011 IEEE 27th International Conference on, IEEE, pp. 75–86.
- [22] J. Cao, K. Hwang, K. Li, A. Y. Zomaya, Optimal multiserver configuration for profit maximization in cloud computing, *IEEE Transactions on Parallel and Distributed Systems* 24 (2013) 1087–1096.
- [23] D. M. Divakaran, M. Gurusamy, M. Sellamuthu, Bandwidth allocation with differential pricing for flexible demands in data center networks, *Computer Networks* 73 (2014) 84–97. *International Journal of Computer Science & Information Technology (IJCSIT)* Vol 10, No 4, August 2018. 38
- [24] J. Zhao, H. Li, C. Wu, Z. Li, Z. Zhang, F. C. Lau, Dynamic pricing and profit maximization for the cloud with geo-distributed data centers, in: *INFOCOM, 2014 Proceedings IEEE*, IEEE, pp. 118–126.
- [25] H. Shen, Rial: Resource intensity aware load balancing in clouds, *IEEE Transactions on Cloud Computing* (2017).
- [26] W. Voorsluys, J. Broberg, S. Venugopal, R. Buyya, Cost of virtual machine live migration in clouds: A performance evaluation, in: *IEEE International Conference on Cloud Computing*, Springer, pp. 254–265.
- [27] Whaiduzzaman, Md. "Performance enhancement framework for cloudlet in mobile cloud computing." PhD diss., Fakulti Sains Komputer dan Teknologi Maklumat, Universiti Malaya, 2016

A NEW CONTEXT-SENSITIVE DECISION MAKING SYSTEM FOR MOBILE CLOUD OFFLOADING

Mustafa Tanrıverdi¹ and M. Ali Akcayol²

¹Institute of Information, Gazi University, Ankara, Turkey ²Department of Computer Engineering, Gazi University, Ankara, Turkey

ABSTRACT

Recently, with the rapid spread use of mobile devices, some problems have begun to emerge. The most important of these are that the mobile devices batteries' life may be short and that these devices may be in some cases. The complex tasks that must be addressed to solve such problems on mobile devices can be transferred to the cloud environment when appropriate conditions are met. The decision to offload to the cloud environment at this stage is very important. In this thesis, a context-aware decision-making system has been developed for offloading to cloud environments. Unlike similar tasks, the processes determined for transfer to the cloud are not run randomly, but rather according to the mobile user's application usage habits. The developed system was implemented in a real environment for one month. According to the results, it was determined that processes transferred to the cloud were completed in less time and consumed less energy.

KEYWORDS

Mobile Cloud Offloading, Mobile Cloud Computing, Context-Aware System, Forecasting, Dynamic Estimation, Energy-Efficiency

For More Details: <http://aircconline.com/ijcsit/V10N3/10318ijcsit05.pdf>

Volume Link: http://airccse.org/journal/ijcsit2018_curr.html

REFERENCES

- [1] S. Hossain, (2014) “What is mobile cloud computing?”, <https://www.ibm.com/blogs/cloudcomputing/2014/01/02/what-is-mobile-cloud-computing>.
- [2] T.Y. Lin, T.A. Lin, C.H. Hsu and C.T. King, (2013) “Context-aware decision engine for mobile cloud offloading”, IEEE Wireless Communications and Networking Conference Workshops (WCNCW), Shanghai, pp. 111–116.
- [3] H. Wu, Q. Wang and K. Wolter, (2013) “Tradeoff between performance improvement and energy saving in mobile cloud offloading systems”, IEEE International Conference on Communications Workshops (ICC), Berlin, pp728–732.
- [4] F. Xia, F. Ding, J. Li, X. Kong, L. T. Yang and J. Ma, (2014) “Phone2Cloud: Exploiting computation offloading for energy saving on smartphones in mobile cloud computing”, Information Systems Frontiers, Vol. 16, pp95–111.
- [5] Y. Hao, M. Chen, L. Hu, M S. Hossain and A. Ghoneim, (2018) “Energy efficient task caching and offloading for mobile edge computing”, IEEE Access, Vol. 6, pp11365–11373.
- [6] S. Kosta, A. Aucinas, P. Hui, R. Mortier and X. Zhang, (2012) “ThinkAir: Dynamic resource allocation and parallel execution in the cloud for mobile code offloading”, INFOCOM 2012 Proceedings IEEE, Orlando, pp945–953.
- [7] B.G. Chun, S. Ihm, P. Maniatis, M. Naik and A. Patti, (2011) “CloneCloud: Elastic execution between mobile device and cloud”, Proceedings of the Sixth Conference on Computer Systems, New York, pp301–314.
- [8] W. T. Su and K. S. Ng, (2013) “Mobile cloud with smart offloading system”, IEEE/CIC International Conference on Communications in China (ICCC), Xi'an, pp680–685.
- [9] AsyncTask, (2016), <https://developer.android.com/reference/android/os/AsyncTask.html>. [10] K. Kumar, J. Liu, Y. H. Lu and B. Bhargava, (2013) “A survey of computation offloading for mobile systems”, Mob. Netw. Appl., Vol. 18, pp129–140.
- [11] B. Zhou, A. V. Dastjerdi, R. N. Calheiros, S. N. Srirama and R. Buyya, (2015) “A context sensitive offloading scheme for mobile cloud computing service”, IEEE 8th International Conference on Cloud Computing, New York, pp869–876.
- [12] C. M. Magurawalage, K. Yang, L. Hu and J. Zhang, (2014) “Energy-efficient and network-aware offloading algorithm for mobile cloud computing”, Comput. Netw., Vol. 74, pp22–33.
- [13] K. H. Lim and B. D. Lee, (2014) “History-based dynamic estimation of energy consumption for mobile applications”, 16th International Conference on Advanced Communication Technology, Pyeongchang, pp714–718.
- [14] R. Aldmour, S. Yousef, M. Yaghi, S. Tapaswi, K. Pattanaik and M. Cole, (2017) “New cloud offloading algorithm for better energy consumption and process time”, Int. J. Syst. Assur. Eng. Manag., Vol. 8, pp730–733.
- [15] L. Li, X. Zhang, K. Liu, F. Jiang and J. Peng, (2018) “An energy-aware task offloading mechanism in multiuser mobile-edge cloud computing”, Mobile Information Systems.

- [16] M. M. Islam, M. A. Razzaque, M. M. Hassan, W. N. Ismail and B. Song, (2017) "Mobile cloudbased big healthcare data processing in smart cities", IEEE Access, Vol. 5, pp11887–11899.
- [17] S. M. A. Karim and J. J. Prevost, (2017) "A machine learning based approach to mobile cloud offloading", Computing Conference, London, pp675–67.
- [18] M. Jia, J. Cao and L. Yang, (2014) "Heuristic offloading of concurrent tasks for computationintensive applications in mobile cloud computing", IEEE Conference on Computer Communications Workshops, Toronto, pp352–357.
- [19] H. Eom, P. S. Juste, R. Figueiredo, O. Tickoo, R. Illikkal and R. Iyer, (2013) "Machine learningbased runtime scheduler for mobile offloading framework", IEEE/ACM 6th International Conference on Utility and Cloud Computing, Dresden, pp17–25.
- [20] H. Flores and S. Srirama, (2013) "Adaptive code offloading for mobile cloud applications: exploiting fuzzy sets and evidence-based learning", Proceeding of the Fourth ACM Workshop on Mobile Cloud Computing and Services, New York, pp9–16.
- [21] M. R. Ra, (2013) "Cloud-enabled mobile sensing systems", PhD Thesis, University of Southern California. International Journal of Computer Science & Information Technology (IJCSIT) Vol 10, No 3, June 2018 89
- [22] Gaurav, N. Kaushik and J. Bhardwaj, (2014) "A computation offloading framework to optimize makespan in mobile cloud computing environment", Int. J. Adv. Comput. Res., Vol. 4, pp442-449.
- [23] R. R. de Oliveira, N. M. S. Schirmer, M. Machry and T. C. Ferreto, (2017) "A transparent code offloading technique for Android devices", 13th International Wireless Communications and Mobile Computing Conference, Valencia, pp1078–1083.
- [24] S. Yang et al., (2013) "Fast dynamic execution offloading for efficient mobile cloud computing", IEEE International Conference on Pervasive Computing and Communications, San Diego, pp 20–28.
- [25] E. Meskar, T. D. Todd, D. Zhao and G. Karakostas, (2017) "Energy aware offloading for competing users on a shared communication channel", IEEE Transactions on Mobile Computing, Vol. 16, pp87- 96.
- [26] S. Yang, D. Kwon, H. Yi, Y. Cho, Y. Kwon and Y. Paek, (2014) "Techniques to minimize state transfer costs for dynamic execution offloading in mobile cloud computing", IEEE Transactions On Mobile Computing, Vol. 13, pp2648-2660.
- [27] SQLite, (2016) <http://www.sqlite.org>.
- [28] M. E. Khoda, M. A. Razzaque, A. Almogren, M. M. Hassan, A. Alamri and A. Alelaiwi, (2016) "Efficient computation offloading decision in mobile cloud computing over 5G network", Mob. Netw. Appl., Vol. 21, pp777–792.
- [29] M. R. Rahimi, N. Venkatasubramanian and A. V. Vasilakos, (2013) "MuSIC: Mobility-aware optimal service allocation in mobile cloud computing", IEEE Sixth International Conference on Cloud Computing, Santa Clara, pp75–82.
- [30] K. Pandi and H. Charaf., (2013) "Performance metrics based mobile resource management", IEEE International Conference on System Science and Engineering, Budapest, pp329-333.

- [31] A. R. S. Nugroho, (2016) “Exploring privacy leakage from the resource usage patterns of mobile apps”, PhD Thesis, University Of Arkansas.
- [32] E. Cuervo et al., (2010) “MAUI: Making smartphones last longer with code offload”, Proceedings of the 8th International Conference on Mobile Systems, Applications, and Services, New York, pp 49– 62.
- [33] C. Chang, (2015) “A Framework for Energy-efficient Mobile Cloud Offloading”, University of Tartu.
- [34] N. I. M. Enzai and M. Tang, (2014) “A taxonomy of computation offloading in mobile cloud computing”, 2nd IEEE International Conference on Mobile Cloud Computing, Services, and Engineering, Oxford, pp19–28.
- [35] K. Korhonen, (2011) “Predicting mobile devices battery life”, Master Thesis, Aalto University Finland.
- [36] J. M. Kang, C. P. Park, S. S. Seo, M. J. Choi, J. W. Hong (2008) “User-Centric predicting for battery lifetime of mobile devices”, Challenges for Next Generation Network Operations and Service Management, pp531-534.
- [37] L. Pu, J. Xu, X. Jin and J. Zhang, (2013) “SmartVirtCloud: virtual cloud assisted application offloading execution at mobile devices' discretion”, Wireless Communications and Networking Conference, Shanghai, pp4398–4403.
- [38] Y. Tao, Y. Zhang, and Y. Ji, (2015) “Efficient computation offloading strategies for mobile cloud computing”, IEEE 29th International Conference on Advanced Information Networking and Applications, Gwangju, pp626–633.
- [39] G. Orsini, D. Bade and W. Lamersdorf, (2018) “CloudAware: Empowering context-aware selfadaptation for mobile applications”, Transactions on Emerging Telecommunications Technologies, Vol. 29.
- [40] S. Yan, C. Shanzhi and X. Xiang, (2018) “MAGA: A mobility-aware computation offloading decision for distributed mobile cloud computing”, IEEE Internet of Things Journal, Vol. 5, pp164-174.

AUTHORS

Dr. Mustafa Tanrıverdi received the Ph.D. degree in Management Information System from Gazi University, Ankara, Turkey, in 2017. He was working in Department of Computer in Gazi University, Turkey until 2007. He has research interest are mobile applications, cloud computing, software development and blockchain.

DISTRIBUTED SCHEME TO AUTHENTICATE DATA STORAGE SECURITY IN CLOUD COMPUTING

B. Rakesh¹ , K. Lalitha¹ , M. Ismail¹ and H. Parveen Sultana²

**¹ Assistant Professor, Department of CSSE, SVEC (Autonomous) ² Associate Professor,
SCOPE, VIT University, Vellore, India**

ABSTRACT

Cloud Computing is the revolution in current generation IT enterprise. Cloud computing displaces database and application software to the large data centres, where the management of services and data may not be predictable, where as the conventional solutions, for IT services are under proper logical, physical and personal controls. This aspect attribute, however comprises different security challenges which have not been well understood. It concentrates on cloud data storage security which has always been an important aspect of quality of service (QOS). In this paper, we designed and simulated an adaptable and efficient scheme to guarantee the correctness of user data stored in the cloud and also with some prominent features. Homomorphic token is used for distributed verification of erasure – coded data. By using this scheme, we can identify misbehaving servers. In spite of past works, our scheme supports effective and secure dynamic operations on data blocks such as data insertion, deletion and modification. In contrast to traditional solutions, where the IT services are under proper physical, logical and personnel controls, cloud computing moves the application software and databases to the large data centres, where the data management and services may not be absolutely truthful. This effective security and performance analysis describes that the proposed scheme is extremely flexible against malicious data modification, convoluted failures and server clouding attacks.

KEYWORDS

Cloud Computing, Cloud Storage Security, Homomorphic token, EC2, S3

For More Details: <http://aircconline.com/ijcsit/V9N6/9617ijcsit06.pdf>

Volume Link: http://airccse.org/journal/ijcsit2017_curr.html

REFERENCES

- [1] Cong Wang, Qian Wang, and Kui Ren, "Ensuring Data Storage Security in Cloud Computing" in Proc. of IWQoS'09, July 2009, pp. 1–9
- [2] G. Ateniese, R. D. Pietro, L. V. Mancini, and G. Tsudik, "Scalable and Efficient Provable Data Possession," Proc. of SecureComm '08, pp. 1–10, 2008.
- [3] Amazon.com, "Amazon Web Services (AWS)," Online at <http://aws.amazon.com>, 2008. [4] A. Juels and J. Burton S. Kaliski, "PORs: Proofs of Retrievability for Large Files," Proc. of CCS '07, pp. 584–597, 2007.
- [5] H. Shacham and B. Waters, "Compact Proofs of Retrievability," Proc. of Asiacrypt '08, Dec. 2008.
- [6] K. D. Bowers, A. Juels, and A. Oprea, "Proofs of Retrievability: Theory and Implementation," Cryptology ePrint Archive, Report 2008/175, 2008, <http://eprint.iacr.org/>.
- [7] K. D. Bowers, A. Juels, and A. Oprea, "HAIL: A High-Availability and Integrity Layer for Cloud Storage," Cryptology ePrint Archive, Report 2008/489, 2008, <http://eprint.iacr.org/>.
- [8] G. Ateniese, R. Burns, R. Curtmola, J. Herring, L. Kissner, Z. Peterson, and D. Song, "Provable Data Possession at Untrusted Stores," Proc. Of CCS '07, pp. 598–609, 2007. International Journal of Computer Science & Information Technology (IJCSIT) Vol 9, No 6, December 2017 66
- [9] R. Curtmola, O. Khan, R. Burns, and G. Ateniese, "MR-PDP: Multiple- Replica Provable Data Possession," Proc. of ICDCS '08, pp. 411–420, 2008.
- [10] M. Lillibridge, S. Elnikety, A. Birrell, M. Burrows, and M. Isard, "A Cooperative Internet Backup Scheme," Proc. of the 2003 USENIX Annual Technical Conference (General Track), pp. 29–41, 2003.
- [11] D. L. G. Filho and P. S. L. M. Barreto, "Demonstrating Data Possession and Uncheatable Data Transfer," Cryptology ePrint Archive, Report 2006/150, 2006, <http://eprint.iacr.org/>. [12] M. A. Shah, M. Baker, J. C. Mogul, and R. Swaminathan, "Auditing to Keep Online Storage Services Honest," Proc. 11th USENIX Workshop on Hot Topics in Operating Systems (HOTOS '07), pp. 1–6, 2007.
- [13] T. S. J. Schwarz and E. L. Miller, "Store, Forget, and Check: Using Algebraic Signatures to Check Remotely Administered Storage," Proc. of ICDCS '06, pp. 12–12, 2006.
- [14] N. Gohring, "Amazon's S3 down for several hours," Online at http://www.pcworld.com/businesscenter/article/142549/amazons_s3_down_for_several_hours.html, 2008.
- [15] K. D. Bowers, A. Juels, and A. Oprea, "HAIL: A High-Availability and Integrity Layer for Cloud Storage," Cryptology ePrint Archive, Report 2008/489, 2008, <http://eprint.iacr.org/>.
- [16] L. Carter and M. Wegman, "Universal Hash Functions," Journal of Computer and System Sciences, vol. 18, no. 2, pp. 143–154, 1979.
- [17] J. Hendricks, G. Ganger, and M. Reiter, "Verifying Distributed Erasurecoded Data," Proc. 26th ACM Symposium on Principles of Distributed Computing, pp. 139–146, 2007.

- [18] J. S. Plank and Y. Ding, "Note: Correction to the 1997 Tutorial on Reed-Solomon Coding," University of Tennessee, Tech. Rep. CS-03- 504, 2003.
- [19] Q. Wang, K. Ren, W. Lou, and Y. Zhang, "Dependable and Secure Sensor Data Storage with Dynamic Integrity Assurance," Proc. of IEEE INFOCOM, 2009.
- [20] R. Curtmola, O. Khan, R. Burns, and G. Ateniese, "MR-PDP: Multiple- Replica Provable Data Possession," Proc. of ICDCS '08, pp. 411–420,2008.
- [21] D. L. G. Filho and P. S. L. M. Barreto, "Demonstrating Data Possession and Unch

MULTILEVEL ANALYSIS OF STUDENT'S FEEDBACK USING MOODLE LOGS IN VIRTUAL CLOUD ENVIRONMENT

Ashok Verma¹, Sumangla Rathore², Santosh Vishwakarma³ and Shubham Goswami⁴

¹Department of Computer Science & Engineering, Sir Padampat Singhania University,
Udaipur, Rajasthan, India

²Department of Computer Science & Engineering, Sir Padampat Singhania University,
Udaipur, Rajasthan, India

³Department of Computer Science & Engineering, Gyan Ganga Institute of Technology &
Sciences, Jabalpur, India

⁴Department of Computer Science & Engineering, Sir Padampat Singhania University,
Udaipur, Rajasthan, India

ABSTRACT

In the current digital era, education system has witness tremendous growth in data storage and efficient retrieval. Many Institutes have very huge databases which may be of terabytes of knowledge and information. The complexity of the data is an important issue as educational data consists of structural as well as non-structural type which includes various text editors like node pad, word, PDF files, images, video, etc. The problem lies in proper storage and correct retrieval of this information. Different types of learning platform like Moodle have implemented to integrate the requirement of educators, administrators and learner. Although this type of platforms are indeed a great support of educators, still mining of the large data is required to uncover various interesting patterns and facts for decision making process for the benefits of the students. In this research work, different data mining classification models are applied to analyse and predict students' feedback based on their Moodle usage data. The models described in this paper surely assist the educators, decision maker, mentors to early engage with the issues as address by students. In this research, real data from a semester has been experimented and evaluated. To achieve the better classification models, discretization and weight adjustment techniques have also been applied as part of the pre – processing steps. Finally, we conclude that for efficient decision making with the student's feedback the classifier model must be appropriate in terms of accuracy and other important evaluation measures. Our experiments also shows that by using weight adjustment techniques like information gain and support vector machines improves the performance of classification models.

KEYWORDS

Educational Data, Educational Data Mining, LMS, Moodle, Feedback system, weight adjustment techniques.

For More Details: <http://aircconline.com/ijcsit/V9N5/9517ijcsit02.pdf>

Volume Link: http://airccse.org/journal/ijcsit2017_curr.html

REFERENCES

- [1] XinGuo, Qing Shi, Danjue Zhang, "A Study on Moodle Virtual Cluster in Cloud Computing", Internet Computing for Engineering and Science (ICICSE), 2013 Seventh International Conference , Publisher: IEEE
- [2] M. Shiraz, S. Abolfazli, Z. Sanaei, and A. Gani, "A study on virtual machine deployment for application outsourcing in mobile cloud computing," The Journal of Supercomputing, vol. 63, March 2013, pp. 946-964, doi:10.1007/s11227-012-0846-y.
- [3] Shiraz M, GaniA (2012) Mobile cloud computing: critical analysis of application deployment in Virtual machines. In: ICICN 2012 IPCSIT, 25–28 February, 2012. IACSIT Press, Singapore
- [4] Chen, Yang, TianyuWo, and Jianxin Li. "An efficient resource management system for on-line virtual cluster provision." Cloud Computing, 2009.CLOUD'09.IEEE International Conference on.IEEE, 2009.
- [5] Ye, Kejiang, et al. "Analyzing and modeling the performance in xen-based virtual cluster environment." High Performance Computing and Communications (HPCC), 2010 12th IEEE International Conference on.IEEE, 2010.
- [8] J. Mostow and J. Beck, Some useful tactics to modify, map and mine data from intelligent tutors, Nat Lang Eng 12 (2006), 195–208, 16
- [9] J. Mostow, J. Beck, H. Cen, A. Cuneo, E. Gouvea, and C. Heiner, An educational data mining tool to browse tutor-student interactions: Time will tell, In: Proceedings of theWorkshop on Educational Data Mining, 2005, pp 15–22.
- [10] L.Dringus and T. Ellis, Using data mining as a strategy for assessing asynchronous discussion forums, Computer & Education 45 (2005), 141–160. Elsevier, Science Direct.
- [11] M.E. Zorrilla, E. Menasalvas, D. Marin, E. Mora, and J. Segovia, Web usage mining project for improving web-based learning sites, In Web Mining Workshop (2005), 1–22.
- [12] O. Zaïane and J. Luo, Web usage mining for a better web-based learning environment, In: Proceedings of the Conference on Advanced Technology for Education, 2001, pp 60–64.
- [13] C. Romero and S. Ventura, Educational data mining: A survey from 1995 to 2005, Expert SystAppl 33 (2007), 135–146.
- [14] C. Romero and S. Ventura, Educational data mining: A review of the state-of- the-art, IEEE Trans Syst Man Cybern C (in press).
- [15] OdedMaimon • LiorRokach, Data Mining and Knowledge Discovery Handbook Second Edition Springer 2010
- [16] Alves G.R., Viegas M.C., Marques M.A., Silva, A.A., Costa-Lobo .C.,Formanski F., Silva, J.B. "Student performance analysis under different Moodle course designs", Interactive Collaborative Learning (ICL), 2012 15th International Conference on DOI: 10.1109/ICL.2012.6402181 Publication Year: 2012, Page(s): 1 - 5
- [17] Daraghmi, E.Y. ; Cheng Hsun Hsiao ; Shyan Ming Yuan "A New Cloud Storage Support and Facebook Enabled Moodle Module" Ubi-Media Computing and Workshops (UMEDIA), 2014 7th International Conference DOI: 10.1109/U-MEDIA.2014.12 Publication Year: 2014 , Page(s): 78 – 83 , 17

- [18] Nagi, K. ;Suesawaluk, P. , “Research analysis of moodle reports to gauge the level of interactivity in elearning courses at Assumption University, Thailand”Computer and Communication Engineering, 2008. ICCCE 2008. International Conference DOI: 10.1109/ICCCE.2008.4580710 Publication Year: 2008, Page(s): 772 - 776,
- [19] Holbl, M. ;Welzer, T. ; Nemec, L. ; Sevcnikar, A. “Student feedback experience and opinion using Moodle” Publication Year: 2011 , Page(s): 1 – 4
- [20] Sael, N. ;Marzak, A. ; Behja, H. Web Usage Mining data preprocessing and multi level analysis on Moodle Computer Systems and Applications (AICCSA), 2013 ACS International Conference Publication Year: 2013 , Page(s): 1 – 7
- [21] Pong-Inwong, C. ; Rungworawut, W. Teaching evaluation using data mining on moodle LMS forum Information Science and Service Science and Data Mining (ISSDM), 2012 6th International Conference on New Trends Publication Year: 2012 , Page(s): 550 – 555 International Journal of Computer Science & Information Technology (IJCSIT) Vol 10, No 5, October 2017 28
- [22] Gil, R. ;Sancristobal, E. ; Diaz, G. ; Castro, M. Biometric verification system in moodle& their analysis in lab exams. International Conference on Computer as a tool. Publication Year: 2011, Page(s): 1 – 4
- [23] Gök, Abdullah, Alec Waterworth, and Philip Shapira. "Use of web mining in studying innovation."Scientometrics 102.1 (2015): 653-671.
- [24] [https://docs.moodle.org/33/en/About Moodle](https://docs.moodle.org/33/en/About_Moodle)
- [25] www.rapidminer.com
- [26] Altujjar, Yasmeen, et al. "Predicting Critical Courses Affecting Students Performance: A Case Study." Procedia Computer Science 82 (2016): 65-71
- [27] Badr, Ghada, et al. "Predicting Students' Performance in University Courses: A Case Study and Tool in KSU Mathematics Department." Procedia Computer Science 82 (2016): 80-89.
- [28] Barba, PG de, Gregor E. Kennedy, and M. D. Ainley. "The role of students' motivation and participation in predicting performance in a MOOC." Journal of Computer Assisted Learning 32.3 (2016): 218-231.
- [29] Pursel, Barton K., et al. "Understanding MOOC students: motivations and behaviours indicative of MOOC completion." Journal of Computer Assisted Learning 32.3 (2016): 202-217.